

CENG 3264

Engineering Design and Project Management

A. Project Self Analysis Paper

Analyzing your project addressing the following:

- **Functional Requirements**
The Biometric Locking Box uses fingerprint authentication to unlock a solenoid-based locking mechanism controlled by an Arduino. It provides secure, user-specific access and automatically relocks after 5 seconds.
- **Constraints**
A major constraint was discovering that the system required a backup power supply to keep the fingerprint sensor and lock functional during outages. Integrating a battery pack while keeping the design simple and low-cost required adjustments to wiring and power distribution.
- **Economic**
The estimated cost was about \$100, but the final cost reached about \$85 after adding a battery pack and materials. Development was planned for three weeks but extended to four due to hardware testing.
- **Environmental**
The system consumes little power and produces minimal waste. Reusing components in future prototypes reduces environmental impact. Using rechargeable batteries helps minimize disposable waste from power sources.
- **Social, Political and Global**
This project demonstrates affordable biometric technology that improves personal security. Globally, similar systems raise privacy concerns regarding storage of biometric data. Our design avoids network use, limiting potential misuse.
- **Ethical**
Ethical considerations focus on protecting stored fingerprint data and ensuring responsible use of biometric systems. The device stores fingerprints locally to maintain user privacy.

- **Relevant Standards**
Although not bound by formal regulations, the design follows general electrical safety and biometric-system principles. Research included examining Arduino hardware voltage standards and sensor communication protocols. All components were used within manufacturer-recommended limits.
- **Health and Safety**
Safety concerns include proper wiring to prevent shorts and safe enclosure of electrical parts. The low-voltage design minimizes shock risk.
- **Manufacturability**
Manufacturing is simple with readily available parts and modular wiring. A custom 3D-printed housing was utilized to protect the fingerprint module.
- **Maintainability**
Maintenance involves cleaning the sensor window.
- **Development**
New tools learned included using the Arduino IDE and fingerprint-sensor libraries for biometric enrollment and verification. Breadboard prototyping improved understanding of circuit integration. These skills enhanced embedded-system design experience.

B. Personal education objectives

Addressing a recognition of the need for, and an ability to engage in **life-long learning**

Throughout this project, I recognized the importance of continuous learning as technology and engineering practices evolve. Working with biometric sensors, embedded systems, and power management showed me that staying current with new tools and methods is essential for effective problem-solving.